CSCI 1851 Homework 1

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1. **Cancer Genomics Dataset**

   There are 801 observations and 5479 features in the Cancer Genomics Dataset. Each feature denotes a gene and the value corresponds to its expression levels measured by the illumina HiSeq platform. I imputed potential missing values using the mean of each feature and removed any duplicated rows.

   The data is high-dimensional as the number of features largely exceeds the number of observations. The COAD (Colon Adenocarcinoma) class is under-represented at 78 instances and the BRCA (Breast Invasive Carcinoma) class is over-represented at 300 instances compared to the other classes. The maximum value for any feature is 20.77, minimum is 0, with a mean of 5.07 and standard deviation of 0.63 (which means feature variances are not very heterogeneous). There is no missing value in this data set. However, the data is very sparse, with 405,815 zero values in the matrix. This means the distribution is likely right-skewed.

2. **Model Performance (Heart Disease + Cancer Genomics)**

   The Gradient Boosted Model performed well on the Cancer Genomics Dataset, achieving a 0.98 accuracy rate, 0.98 precision rate, 0.98 recall rate, 0.98 f-1 score, and 1.0 AUC-ROC on the unscaled data. This indicates that the model is great at detecting the correct class, and is equally likely to make a Type I or Type II error.

   The Gradient Boosted Model achieved moderate performance on the Heart dataset. It reached a 0.77 accuracy rate, 0.76 precision rate, and 0.84 recall rate on the unscaled data after cross-validation. This indicates that the model is slightly more likely to make a Type I error than a Type II error (adjusting the number of trees may influence the voting decision threshold).

   Surprisingly, the heart dataset was harder to classify. Perhaps the different cancer types contain distinct marker genes that allow easy clustering, while for heart disease there are a lot of overlapping features between positive and negative cases. Here are the Cross-Validate result (the metrics for the cancer genomics are weighted to more accurately reflect performance):

Table 1: Heart CV Results

	Accuracy	**Precision**	**Recall**	**F1**	**ROC-AUC**
Mean	0.7719	0.7618	0.8476	0.8006	0.8743
Std	0.0256	0.0256	0.0681	0.0279	0.0339

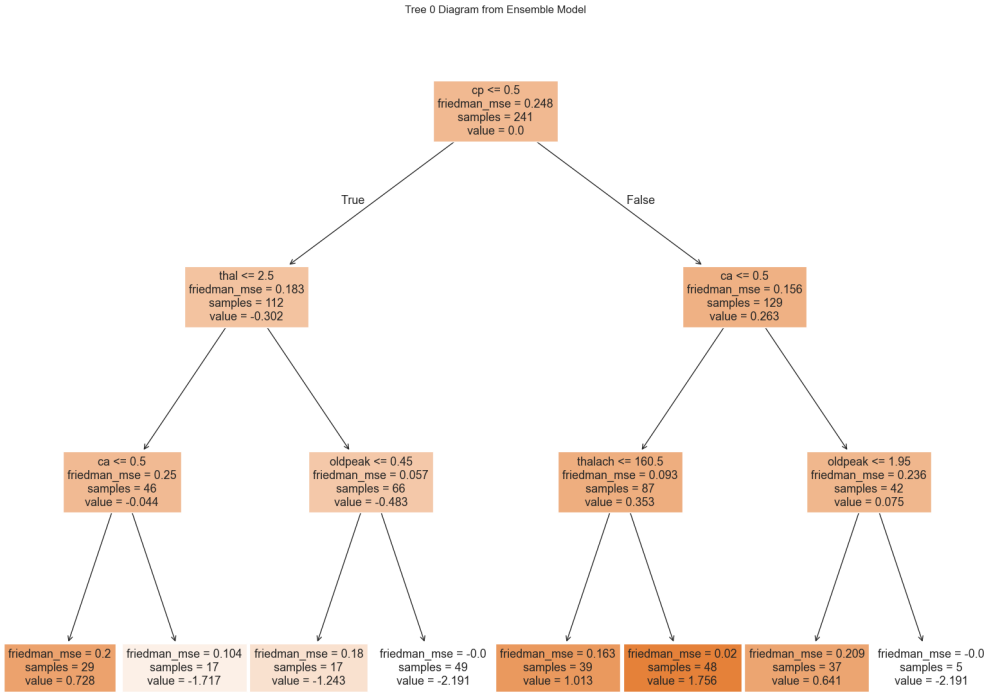

Figure 1: heart Disease Tree Diagram

Table 2: Cancer Genomics CV Results

	Accuracy	**Precision (W)**	**Recall (W)**	**F1 (W)**	**ROC-AUC (W)**
Mean	0.9875	0.9877	0.9875	0.9874	0.9983
Std	0.0106	0.0105	0.0106	0.0107	0.0031

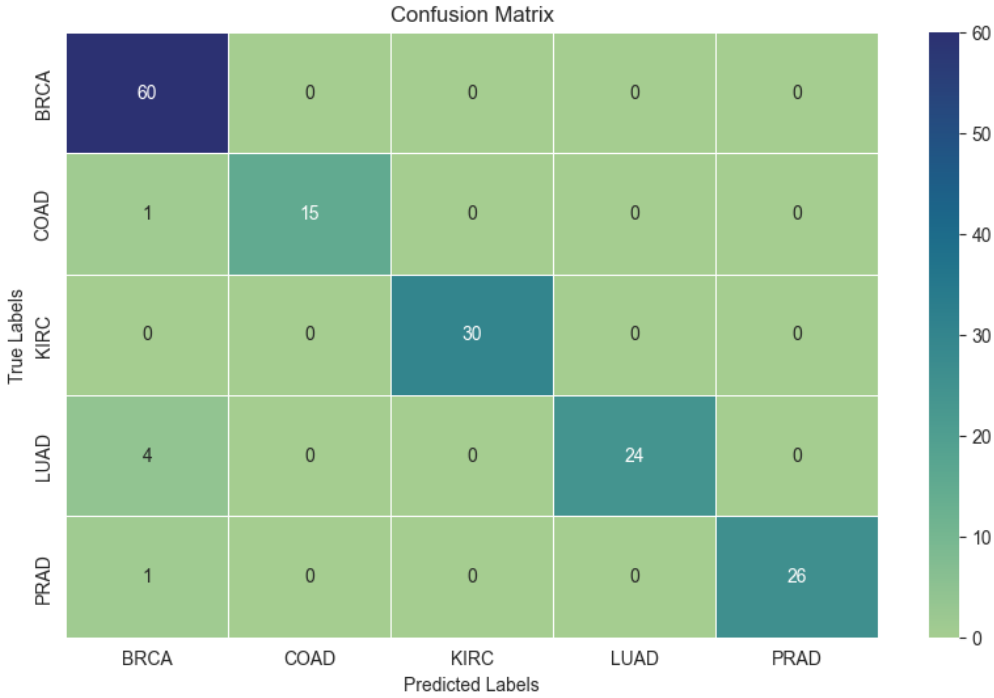

Figure 2: Cancer Genomics Confusion Matrix

3. **Hyperparameter Tuning** (**Gradient Boosting**)

   For the heart data set, I explored tuning the parameters max depth, n estimators, and learning rate, with ranges (2,3,5,10), (10,30,50), and (0.05,0.1,0.2), respectively. Ranked by AUC-ROC, the best results were achieved by a max depth of 2, an n estimator of 50, and a learning rate of 0.05. The model favored shallower trees and slower learning rates, which is reasonable as we know the Heart Disease data is harder to classify, and the model seems to be converging more carefully by requiring more estimators.

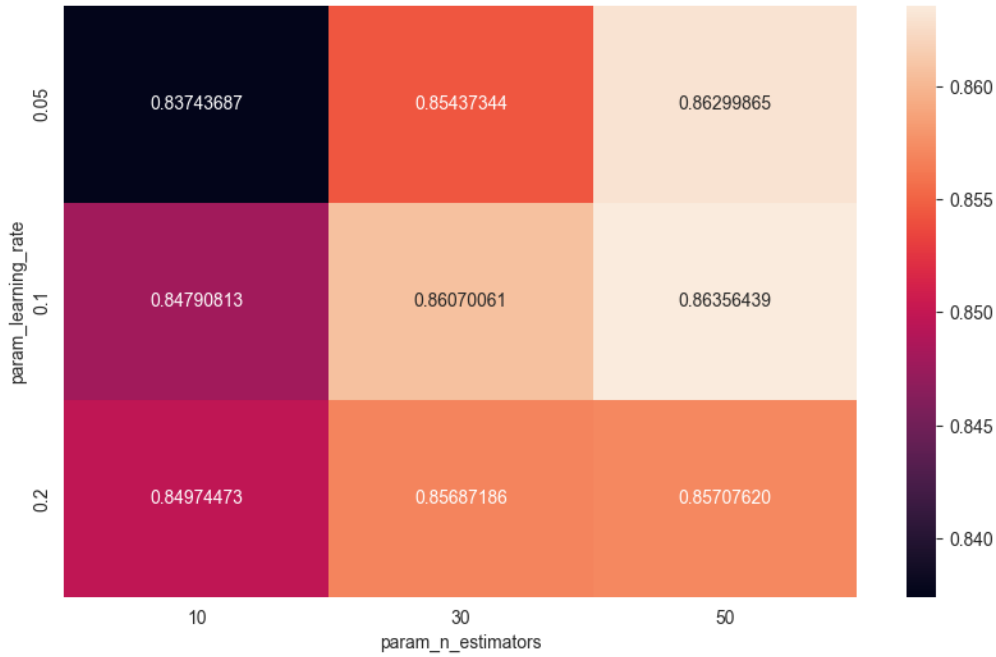

Figure 3: Heart Disease Grid Search Heatmap

For the Genomics data set, I explored using more tuning parameters including max depth (2,5,10), n estimators (30,50,70), learning rate (0.1,0.2), subsample (0.5,1), and minimum sample leaf (2,5) to further adjust over/underfitting behavior. I used a Cross-Validation of 3 and found the best results (accuracy = 0.99, precision = 0.99, recall = 0.99, f1 = 0.99, AUC-ROC = 1) were achieved by a max depth of 2, learning rate of 0.2, n estimators of 70, subsample of 0.5, and minimum sample leaf of 5. It is interesting to note that the best parameters optimized for a large ensemble and a more shallow tree structure with only 50% training data used for each tree. The learning rate is also quite high, which means the signal is quite strong and allows the model to converge quickly. This set up is less prone to overfitting, and the structure is suited for data sets that can be separated easily.

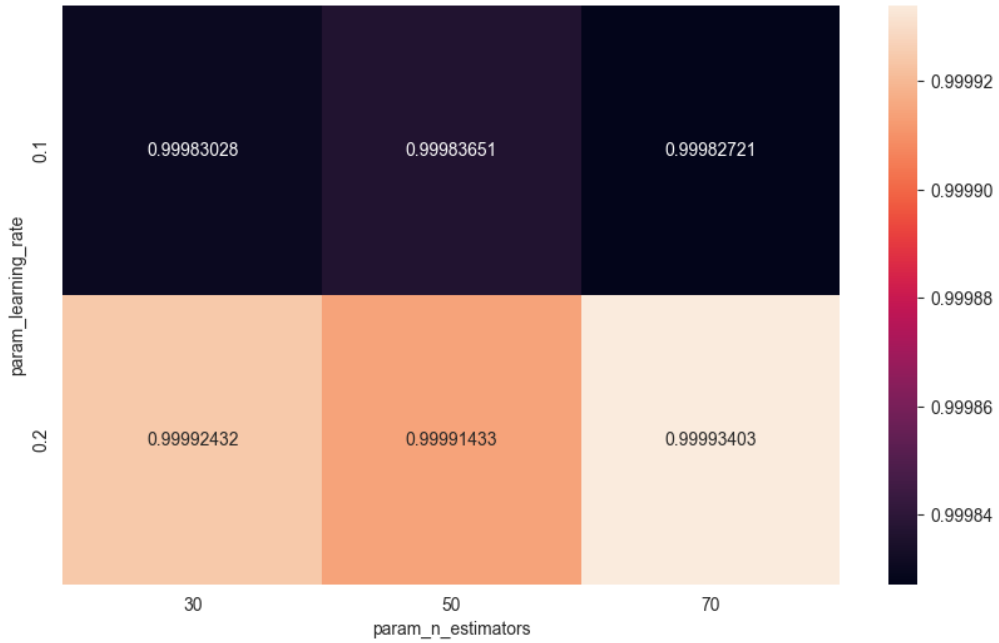

Figure 4: Cancer Genomics Grid Search Heatmap

4. **Feature Importance and Interpretation**

   We proceed to fit the model using the parameters we obtained from the hyperparameter tuning step. The top 5 features of the Gradient Boosted Model trained on the Cancer Genomics dataset are gene 7964 (0.132), gene 203 (0.097), gene 14503 (0.091), gene 15896 (0.087), and gene 17801 (0.083). Interestingly, the feature importance tapers off gradually, and after the top 20 features, the rest of the genes all have a near-zero importance measure. This could be possible if the genetic profile of the different cancer types are very distinct from each other (e.g., certain genes are only expressed in COAD patients), which would allow the model to easily differentiate them using only a few genes.

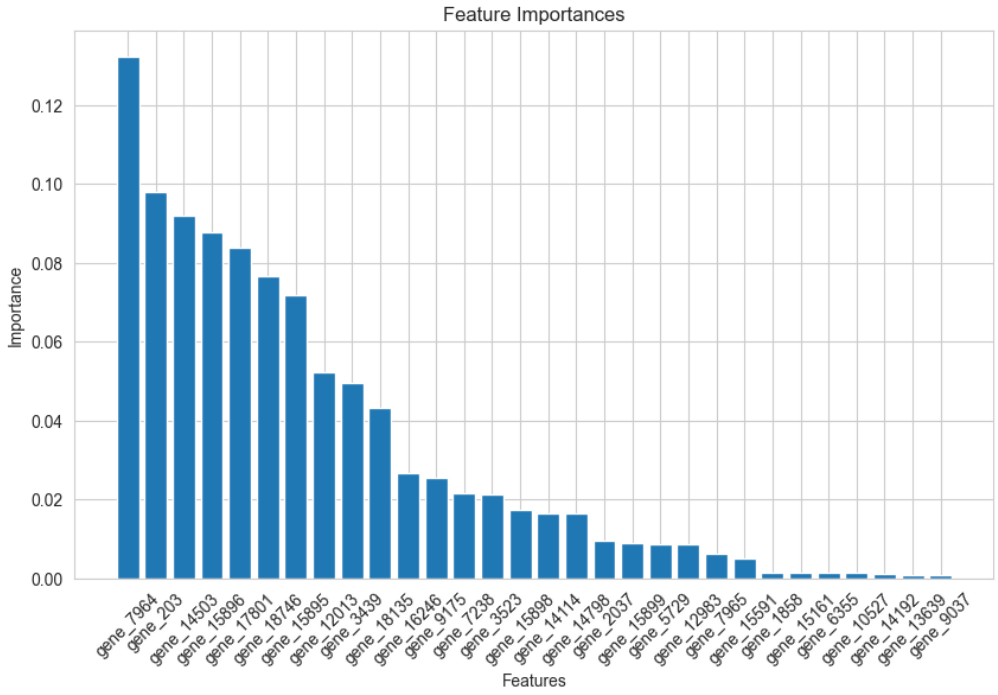

Figure 5: Cancer Genomics Top 30 Feature Importance

For the Heart Disease dataset, the top 5 features are Chest Pain Type (0.32), thal (0.11), Number of Major Vessels under fluoroscopy (0.10), Cholesterol (0.09), and oldpeak (0.08). These features overlap with the top features (interpreted using coefficients) identified by the Logistics Regression model in HW1. This seems to suggest that some of the features, such as Chest Pain Type, is both good for making classification ("splitting") decisions for Heart Disease status and positively correlate with Heart Disease.

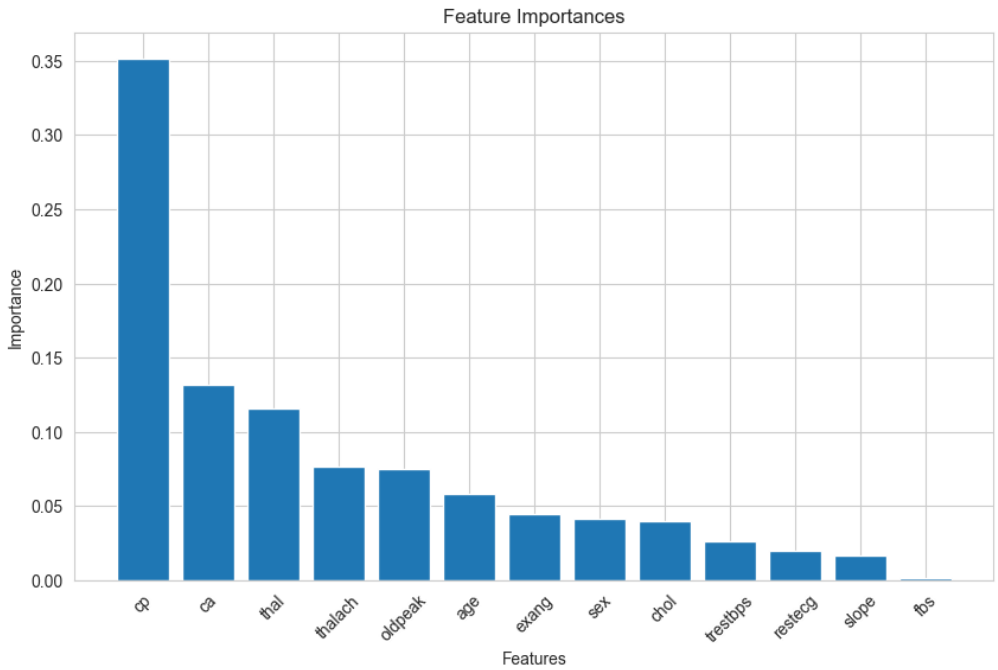

Figure 6: Heart Disease Top 10 Feature Importance

5. **Comparison to HW1**

   I instantiated a Logistic Regression model using a similar model set up from the HW1 (Heart Disease Classifer). I used the default method of "One vs Rest", which tasks the model to train separate binary classifers for each class and to make predictions based on the probability of a sample being within that class or not. Interestingly, the Logistics Regression model trained on the full feature set yielded better results than the Gradient Boosted Classifer model. The logistics regression model achieved an accuracy of 0.99, precision of 0.99, and recall of 0.99.

   Training the model on the top 5 most important features identified by the Gradient Boosted Classifier, we obtain the results of 0.98 for accuracy, 0.98 for precision, 0.98 for recall, 0.98 for f-1, and 1 for AUC-ROC. This set of results is lower than the results from using the full feature set, but it suggests that most of the signal is captured by the top 5 important genes. Using the top 20 most important features returns 1 for all evaluation metrics, which could suggest that the majority of the genes contribute noise and hurt the model, but the exceedingly high scores means that data leakage might have occurred during the training process.

6. **Discussion**

   Our Gradient Boosted Classifier model achieved good performance on both the Heart Disease and Cancer Genomics data sets. It is non-parametric, does not assume the data follows any distribution, and can handle class imbalance through weighting configuration. However, as our results have shown, it can be sensitive to hyper-parameters (e.g., max depth, number of estimators) and therefore could lead to overfitting if the tree becomes too complex through including too many estimators or too many layers. Our data set is high dimensional with a lot of sparsity, which means it might be very noisy. The curse of dimensionality can make it

harder for the model to learn the underlying structure and make accurate predictions if the testing data appears in a feature space with no training examples. The model also works well when data set can be clustered cleanly, but has trouble when there are many features with overlapping class distributions (as we see from the Heart Disease vs Cancer Genomics data set). Yet the near-perfect performance on the Cancer Genomics data set could indicate data leakage in the training process (e.g., important features inferred from the test set was used in the training set as well). Another limitation of the model is the computational constraint, as I was only able to tune and test a subset of hyperparameters in grid search due to having limited computational power.